

Vision and Text Integration

Naeemullah Khan

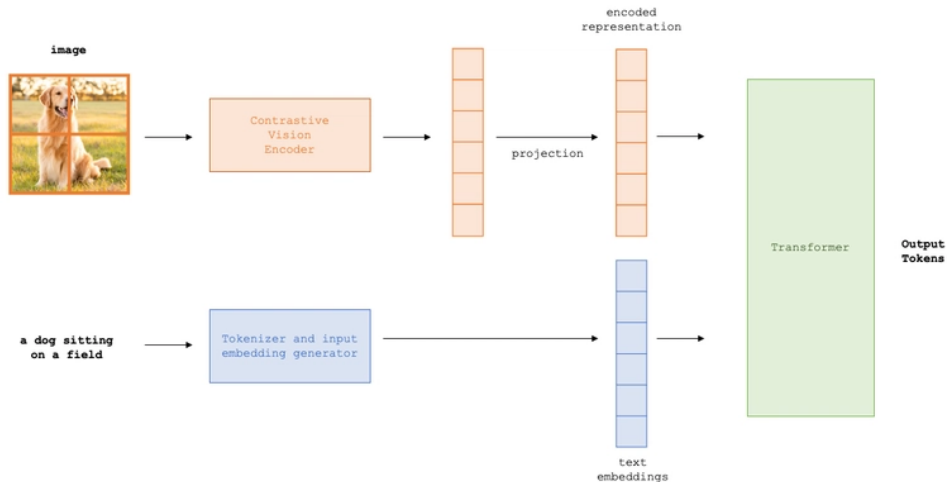
naeemullah.khan@kaust.edu.sa



جامعة الملك عبد الله
للعلوم والتقنية
King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

August 9, 2026



[Source: S. Debnath, "Agentic RAG with ColPali", Medium, 2025,
https://medium.com/@suman.d_16555/agentic-rag-with-colpali-3982e69a770d]

1. Motivation
2. Learning Outcomes
3. Image and Video Captioning
 1. Image Captioning Pipeline
 2. Show and Tell
 3. Video Captioning
 4. Challenges in Captioning
 5. Recent Advances and Datasets
4. Multimodal Models
 1. Multimodal Representation Learning
 2. Joint Embedding Spaces for Image-Text Pairs
 3. Contrastive Learning for Multimodal Models

- 4. CLIP: Contrastive Language–Image Pretraining
- 5. Text-to-Image Generation Techniques
 - 1. GAN-based Approaches
 - 2. Diffusion-based Approaches
 - 3. Measuring Text-to-Image Quality
- 6. Text-Conditioned Generation
 - 1. Overview of Text-Conditioned Generation
 - 2. Prompt Engineering
 - 3. Challenges in Conditioning
- 7. Classifier-Free Guidance (CFG)
- 8. Summary
- 9. References

- ▶ The human brain naturally integrates vision and language (e.g., describing scenes, reading captions).
- ▶ This inspires artificial systems to process and relate visual and textual information.
- ▶ Recent applications:
 - Image captioning
 - Visual question answering (VQA)
 - Generative AI

- ▶ Joint vision-language understanding is crucial for intelligent systems.
- ▶ Foundation for:
 - Autonomous agents
 - Assistive technologies
 - Content creation tools

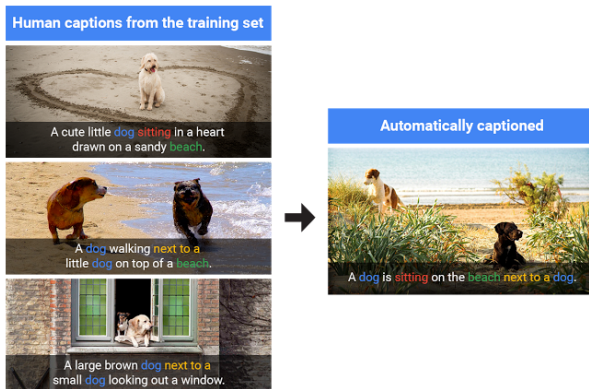
By the end of this session, learners will be able to:

- ▶ Understand how vision and text can be jointly modeled.
- ▶ Explore state-of-the-art models such as CLIP and diffusion-based text-to-image (T2I) methods.
- ▶ Implement and evaluate image and video captioning systems.
- ▶ Grasp classifier-free guidance and text-conditioned generation techniques.
- ▶ Analyze the limitations of current approaches and propose future research directions.

Vision and Text Integration: **Image and Video Captioning**

What is Captioning?

- **Definition:** Generating a descriptive text from an image or a video.



Source: Google Research Blog, "Show and Tell: image captioning open sourced in TensorFlow", 2016.

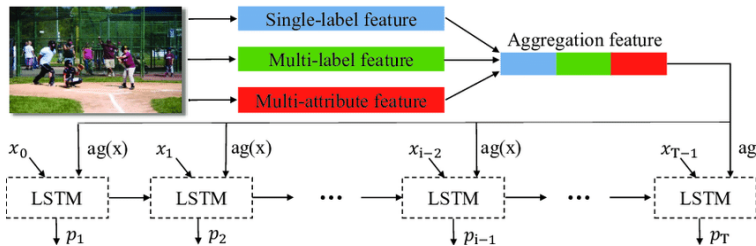
► Challenges:

- Semantics
- Scene understanding
- Temporal context

► Applications:

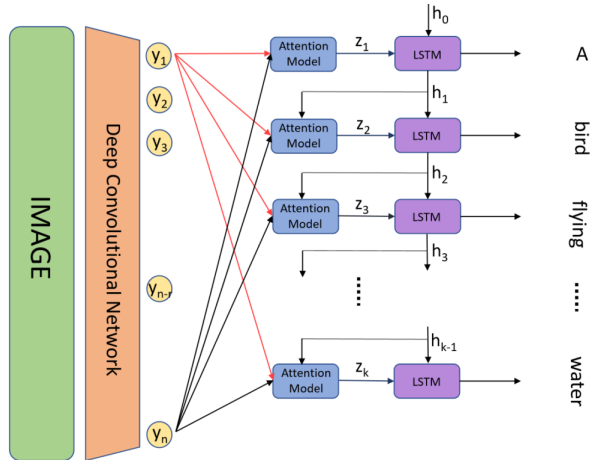
- Accessibility
- Content indexing
- Social media
- Surveillance

- **Feature Extraction:** Use a CNN (e.g., ResNet) to extract visual features from the image.
- **Language Modeling:** Use an RNN, LSTM, or Transformer to generate a textual description based on the extracted features.
- **Encoder-Decoder Architecture:** The CNN acts as the encoder and the RNN/LSTM/Transformer as the decoder.



Convnet features fed to the LSTM at every step (Liu, Guo & Lew, MMM 2017).

Image Captioning Pipeline (cont.)

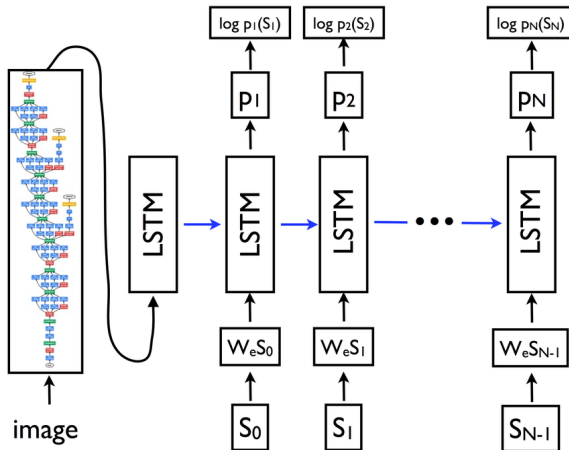


Attention-based image captioning (Show, Attend and Tell, Xu et al., ICML 2015)

Example: Show and Tell (Vinyals et al., 2015)

- ▶ One of the first successful end-to-end image captioning models.
- ▶ Uses a CNN (Inception) as encoder and an LSTM as decoder.
- ▶ Trained to maximize the likelihood of the correct caption given the image.

Example: Show and Tell (Vinyals et al., 2015) (cont.)



Show and Tell model architecture (Vinyals et al., 2015).

Example: Show and Tell (Vinyals et al., 2015) (cont.)

Helping visually impaired people better understand the content of images on the web.

- Pascal dataset
- Flickr30k dataset
- SBU dataset
- COCO dataset

$I \rightarrow$ input image

$S = \{S_1, S_2, \dots\} \rightarrow$ target sequence of words

$p(S|I) \rightarrow$ likelihood

S_t comes from a given dictionary

NIC: Neural Image Caption

$$\theta^* = \arg \max_{\theta} \sum_{(I, S)} \log p(S|I; \theta)$$

$$\log p(S|I) = \sum_{t=0}^N \log p(S_t|I, S_0, \dots, S_{t-1})$$

model with an RNN
LSTM $h_{t+1} = f(h_t, x_t)$

CNN input (image & words)

$$i_t = \sigma(W_{ix}x_t + W_{im}m_{t-1})$$

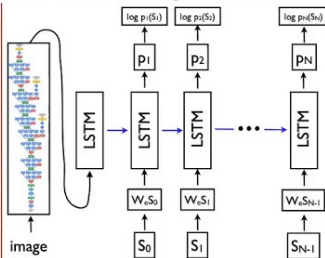
$$f_t = \sigma(W_{fx}x_t + W_{fm}m_{t-1})$$

$$o_t = \sigma(W_{ox}x_t + W_{om}m_{t-1})$$

$$c_t = f_t \odot c_{t-1} + i_t \odot h(W_{cx}x_t + W_{cm}m_{t-1})$$

$$m_t = o_t \odot c_t$$

$$p_{t+1} = \text{Softmax}(m_t)$$



$$x_{-1} = \text{CNN}(I)$$

$$x_t = W_e S_t, \quad t \in \{0 \dots N-1\}$$

$$p_{t+1} = \text{LSTM}(x_t), \quad t \in \{0 \dots N-1\}$$

$S_0 \rightarrow$ start symbol

$S_N \rightarrow$ end symbol

$S_t \rightarrow$ one-hot-vector

$W_e \rightarrow$ word embeddings

$$L(I, S) = - \sum_{t=1}^N \log p_t(S_t)$$

BeamSearch: Iteratively consider the set of the k best sentences up to time t as candidates to generate sentences of size $t+1$, and keep only the resulting best k of them. (beam size: $k = 20$)



Vinyals, Oriol, et al. "Show and tell: A neural image caption generator." Proceedings of the IEEE conference on computer vision and pattern recognition. 2015.

Example: Show and Tell (Vinyals et al., 2015) (cont.)

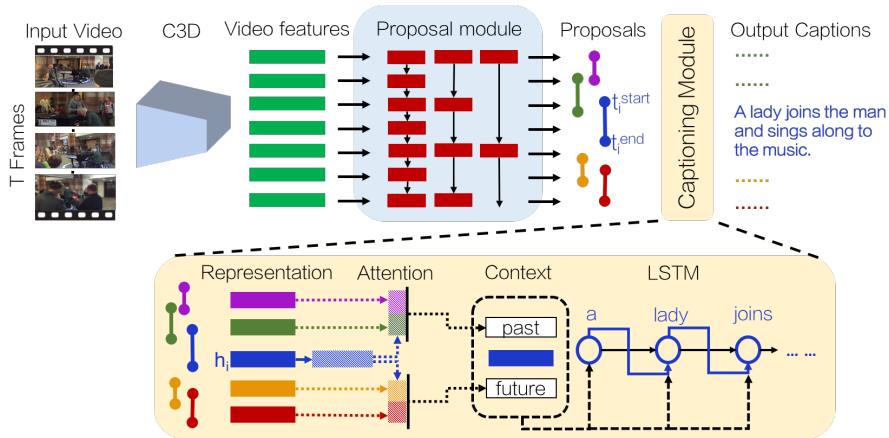


Figure 5. A selection of evaluation results, grouped by human rating.

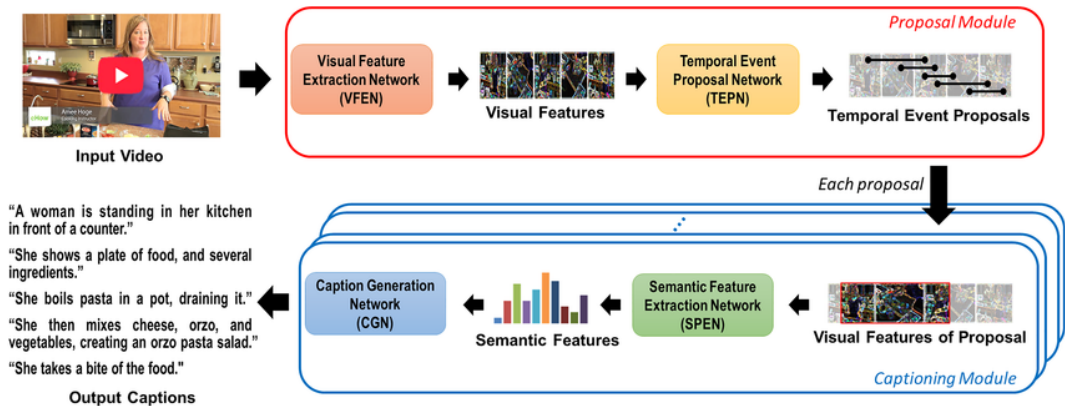
A selection of evaluation results, grouped by human rating (Vinyals et al., CVPR 2015, Fig. 5).

- ▶ **Video Captioning:** Generating descriptive text for a sequence of video frames.
- ▶ **Temporal Sequence Modeling:** Adds modeling of temporal dependencies between frames, in addition to spatial features.
- ▶ **Common Architecture:**
 - CNN for frame-level feature extraction.
 - RNN (e.g., LSTM/GRU) for modeling temporal sequence.
 - Attention mechanisms to focus on relevant frames or regions.
- ▶ **Recent Advances:**
 - Transformer-based models for parallel sequence modeling.
 - Memory networks for capturing long-term dependencies.

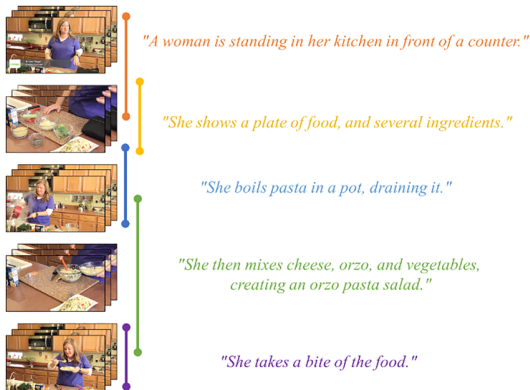
Video Captioning (cont.)



Dense-Captioning Events in Videos (Krishna et al., ICCV 2017)



DVC-Net model for dense video captioning (Lee & Kim, *IET Computer Vision* 15(1), 2021, Fig. 2).

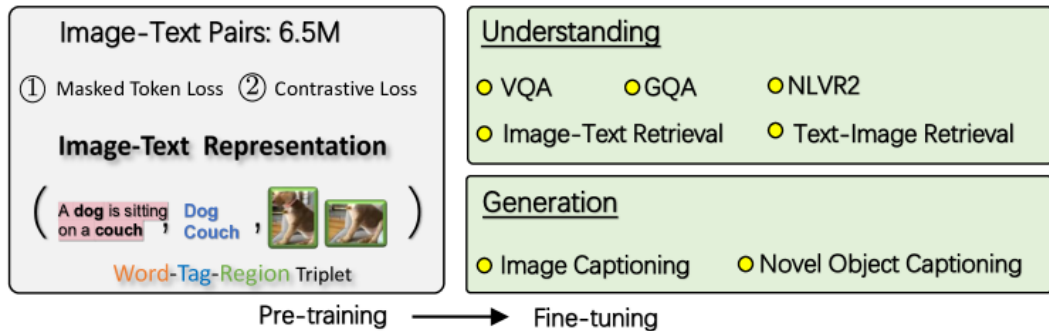


One video, five time-localised events described in natural language (Lee & Kim, *IET Computer Vision* 15(1), 2021).

- ▶ **Ambiguity:** Multiple valid captions for the same image or video.
- ▶ **Context Understanding:** Captions need to capture context and semantics accurately.
- ▶ **Temporal Dynamics:** In video captioning, capturing the dynamics of actions and events over time is crucial.
- ▶ **Data Scarcity:** High-quality annotated datasets for training captioning models are limited.
- ▶ **Evaluation Metrics:** Evaluating caption quality is subjective. The usual scores compare a generated caption against human references: BLEU on n-gram precision, METEOR on stem and synonym matches weighted towards recall, CIDEr on TF-IDF weighted n-gram similarity. None of them tracks human judgment closely.

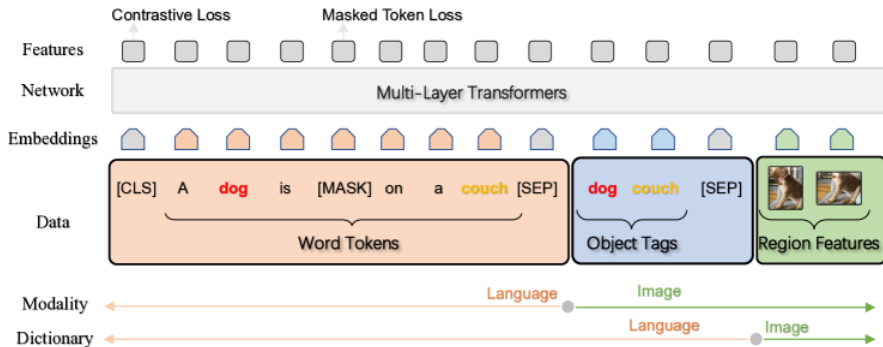
Image Captioning Advances:

Oscar: Object-Semantics Aligned Pre-training for Vision-Language Tasks.



Oscar pre-training and fine-tuning pipeline (Li et al., ECCV 2020, Fig. 1).

Oscar: Object tags anchor image regions to the words that name them.



Word tokens, object tags and region features share one Transformer (Li et al., ECCV 2020, Fig. 3).

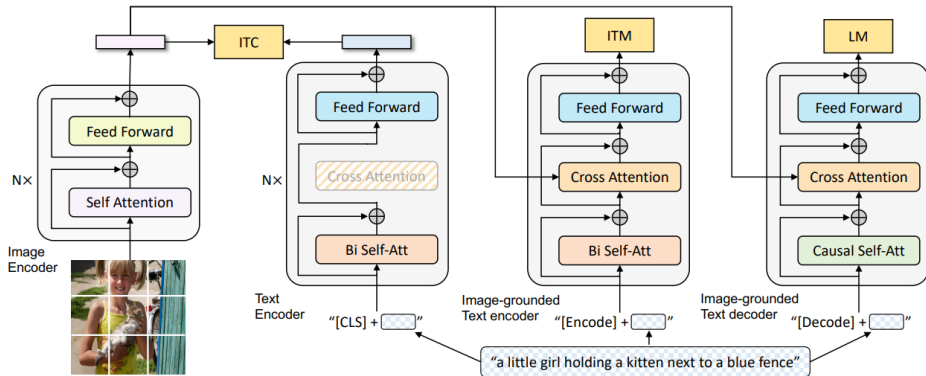
VinVL: A stronger object detector supplies richer region features than the bottom-up detector Oscar used.



Regions found by the previous detector (left) and by the VinVL detector (right) (Zhang et al., CVPR 2021, Fig. 1).

Recent Advances in Captioning (cont.)

BLIP: Bootstrapping Language-Image Pre-training for unified vision-language understanding and generation.



BLIP (Li et al., ICML 2022).

Image Captioning Advances (continued):

- ▶ **ClipCap:** A small mapping network turns a CLIP image embedding into a prefix for a GPT-2 language model, which writes the caption (Mokady et al., 2021).

Video Captioning Advances:

- ▶ **MARN:** Memory-Attended Recurrent Network for video captioning (Pei et al., CVPR 2019).
- ▶ **STG-KD:** Spatio-Temporal Graph for Video Captioning with Knowledge Distillation (Pan et al., CVPR 2020).

Key Datasets

- ▶ **MSCOCO:** Large-scale dataset for image captioning.
- ▶ **ActivityNet Captions:** Benchmark for dense video captioning.
- ▶ **YouCook2:** Large-scale dataset of instructional cooking videos with captions.

Vision and Text Integration: **Multimodal Models**

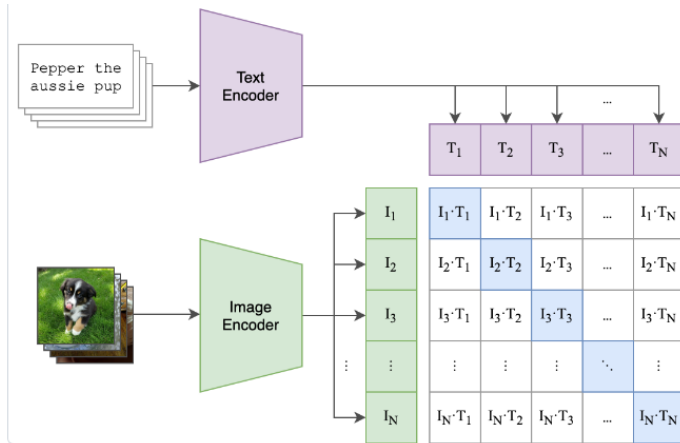
- ▶ **Definition:** Multimodal models place visual and textual information in a shared representation, so an image and the text describing it can be compared, retrieved, or generated from one another.
- ▶ **Challenges:**
 - Ambiguity in visual content
 - Context understanding
 - Temporal dynamics in videos

► Applications:

- Image captioning
- Video captioning
- Cross-modal retrieval
- Zero-shot classification
- Assistive technologies (e.g., for visually impaired)

- ▶ **Goal:** Learn shared representations for different modalities (e.g., images and text).
- ▶ **Approach:** Map both images and text into a joint embedding space.
- ▶ **Benefit:** Enables comparison and retrieval across modalities.

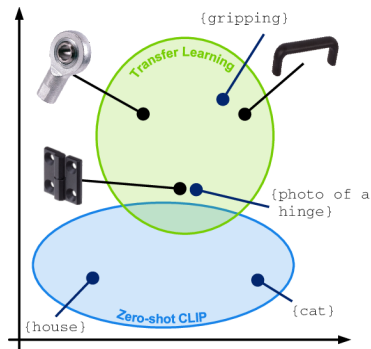
Multimodal Representation Learning (cont.)



Multimodal representation learning: mapping images and text into a joint embedding space (Radford et al., ICML 2021, Fig. 1, panel 1).

- ▶ Images and their corresponding captions are projected into a common vector space.
- ▶ Matching image-text pairs are close together; non-matching pairs are far apart.
- ▶ Used in tasks like cross-modal retrieval and caption generation.

Joint Embedding Spaces for Image-Text Pairs (cont.)



Joint embedding space for image-text pairs: matching pairs are close, non-matching pairs are distant.

[Figure source: ResearchGate, <https://www.researchgate.net/figure/>

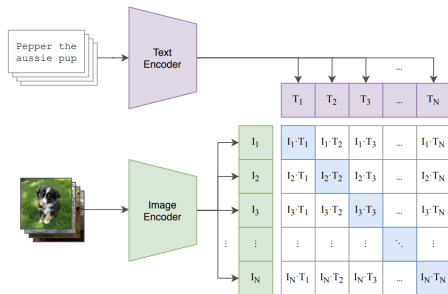
[Joint-embedding-space-of-text-and-image-representations-conceptually-similar-texts-and_fig2_381470711](https://www.researchgate.net/figure/Joint-embedding-space-of-text-and-image-representations-conceptually-similar-texts-and_fig2_381470711)]

- ▶ **Contrastive Objective:** Pull matching image-text pairs together, push non-matching pairs apart.
- ▶ **Popular Methods:** CLIP, ALIGN, etc.
- ▶ **Loss Function:** Often uses InfoNCE or similar contrastive losses.

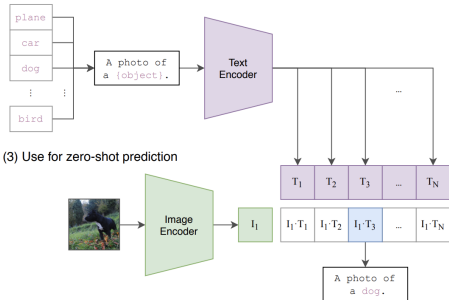
- ▶ **Overview:** CLIP is a multimodal model trained to connect images and text using contrastive learning.
- ▶ **Training Data:** 400 million image-text pairs collected from the internet.
- ▶ **Architecture:**
 - Two separate encoders: a text Transformer, and an image encoder that is either a modified ResNet or a Vision Transformer.
 - Both modalities are mapped into a shared embedding space.
- ▶ **Contrastive Loss:** Uses dot-product similarity to align matching image-text pairs and separate non-matching pairs.
- ▶ **Zero-shot Classification:** Enables classification of images without task-specific training by comparing image embeddings to text prompt embeddings.

CLIP (Contrastive Language–Image Pretraining) (cont.)

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

CLIP: separate encoders for images and text, mapping both into a shared embedding space (Radford et al., ICML 2021, Fig. 1).

Why CLIP Matters:

- ▶ **Open-vocabulary vision models:** CLIP recognises concepts beyond a fixed label set, as long as they can be named in text.
- ▶ **Robust to distribution shift:** Performs well on images from domains not seen during training.
- ▶ **No task-specific training required:** A new task is set up by writing text prompts for its classes, not by collecting labels.

Limitations of CLIP:

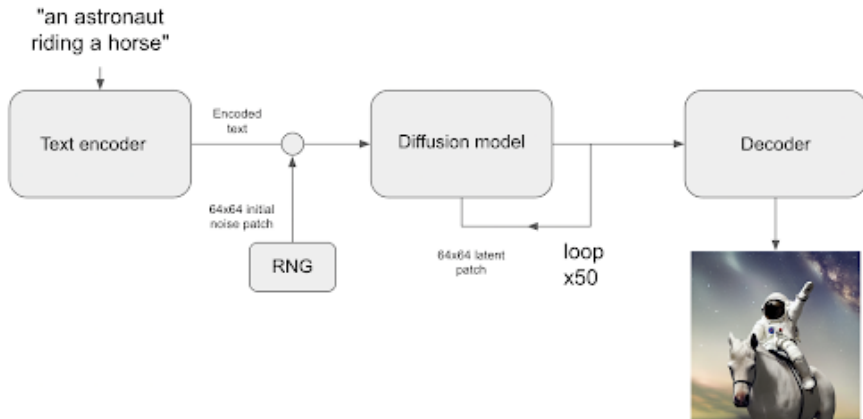
- ▶ **Struggles with fine-grained details:** May fail to distinguish subtle differences between visually similar objects.
- ▶ **Lacks true reasoning abilities:** Cannot perform complex reasoning or understand relationships beyond surface-level associations.
- ▶ **Prone to dataset biases:** Inherits biases present in the large-scale web data used for training.

Text-to-Image Generation Techniques

- ▶ Text-to-image generation involves creating images from textual descriptions.
- ▶ Requires alignment between vision and language features.
- ▶ Key techniques include:
 - Diffusion models
 - Generative adversarial networks (GANs)
 - Variational autoencoders (VAEs)
- ▶ Applications:
 - Art and design
 - Content creation
 - Entertainment
 - Virtual environments
 - Accessibility

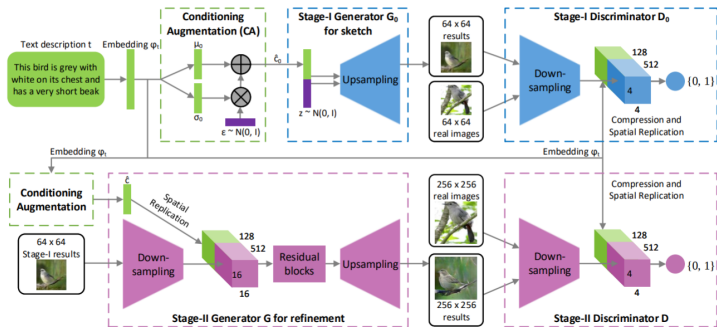
- ▶ Key challenges:
 - Ensuring high fidelity to the text description.
 - Maintaining diversity in generated images.
 - Handling complex and abstract descriptions.

Text-to-Image Generation (cont.)



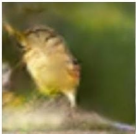
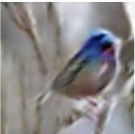
[Source: https://miro.medium.com/v2/resize:fit:1028/0*26may-706_HoRFD7]

StackGAN (Zhang et al., ICCV 2017): Generates images in two stages. Stage-I sketches the primitive shape and colours at 64×64 ; Stage-II re-reads the text and refines to 256×256 .



StackGAN two-stage architecture with Conditioning Augmentation (Zhang et al., ICCV 2017).

GAN-based Approaches (cont.)

Text description	This bird is blue with white and has a very short beak	This bird has wings that are brown and has a yellow belly	A white bird with a black crown and yellow beak	This bird is white, black, and brown in color, with a brown beak	The bird has small beak, with reddish brown crown and gray belly	This is a small, black bird with a white breast and white on the wingbars.	This bird is white black and yellow in color, with a short black beak
Stage-I images							
Stage-II images							

Stage-I and Stage-II samples (Zhang et al., ICCV 2017).

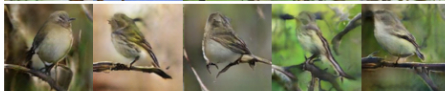
This small blue bird has a short pointy beak and brown on its wings



This bird is completely red with black wings and pointy beak



A small sized bird that has a cream belly and a short pointed bill



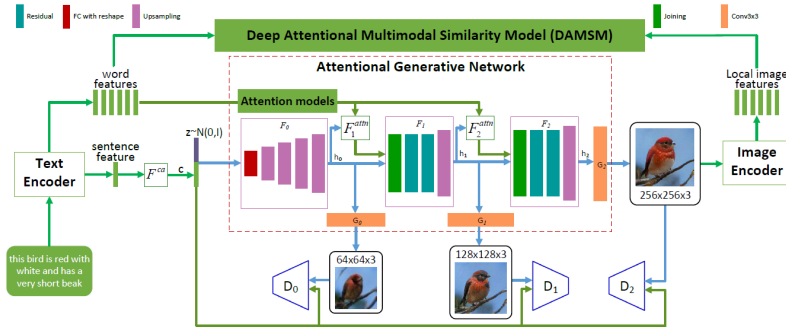
A small bird with a black head and wings and features grey wings



Figure 7. Birds with different poses and viewpoints generated with the same input text embedding by our StackGAN. The noise vector z and text embedding are fixed for each row.

Birds generated with different poses from the same text (Zhang et al., ICCV 2017).

AttnGAN (Xu et al., CVPR 2018): Attention lets each image sub-region attend to the relevant words. A deep attentional multimodal similarity model (DAMSM) supplies a fine-grained image-text matching loss.



AttnGAN attentional generative network with the DAMSM loss (Xu et al., CVPR 2018).

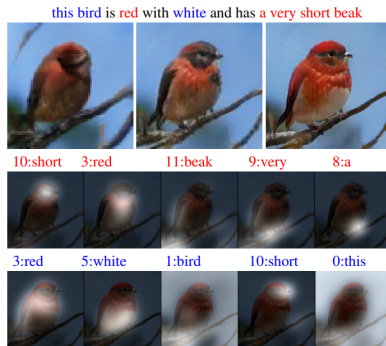


Figure 1. Example results of the proposed AttnGAN. The first row gives the low-to-high resolution images generated by G_0 , G_1 and G_2 of the AttnGAN; the second and third row shows the top-5 most attended words by F_1^{attn} and F_2^{attn} of the AttnGAN, respectively. Here, images of G_0 and G_1 are bilinearly upsampled to have the same size as that of G_2 for better visualization.

Multi-stage AttnGAN results with word-level attention maps (Xu et al., CVPR 2018).

Stage-wise Generation:

- ▶ Images are generated progressively, refining details at each stage.

Limitations:

- ▶ Training instability
- ▶ Mode collapse (lack of diversity in generated images)

- ▶ Diffusion models generate images by iteratively denoising random noise, conditioned on text prompts.
- ▶ Notable models:
 - DALL•E 2
 - Imagen
 - Stable Diffusion
- ▶ **Key idea:** Learn to reverse a gradual noising process, producing high-quality images from pure noise.
- ▶ **Advantages:**
 - Superior image quality and diversity compared to GANs and VAEs
 - Better alignment with textual descriptions

DALL•E 2 (unCLIP; Ramesh et al., 2022): A *prior* maps the CLIP text embedding to a CLIP *image* embedding; a diffusion *decoder* turns that into pixels. CLIP stays frozen throughout.

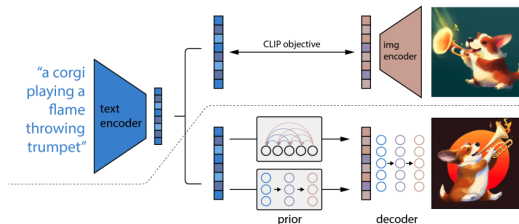


Figure 2: A high-level overview of unCLIP. Above the dotted line, we depict the CLIP training process, through which we learn a joint representation space for text and images. Below the dotted line, we depict our text-to-image generation process: a CLIP text embedding is first fed to an autoregressive or diffusion prior to produce an image embedding, and then this embedding is used to condition a diffusion decoder which produces a final image. Note that the CLIP model is frozen during training of the prior and decoder.

DALL•E 2 / unCLIP overview (Ramesh et al., 2022, Fig. 2).



[Source: https://miro.medium.com/v2/resize:fit:1400/0*TUVPZ9ZCD4q8QdUB]

Imagen: A diffusion model by Google that generates high-quality images from text, focusing on photorealism and fine details.

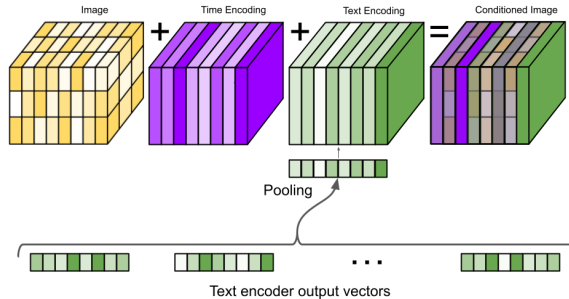
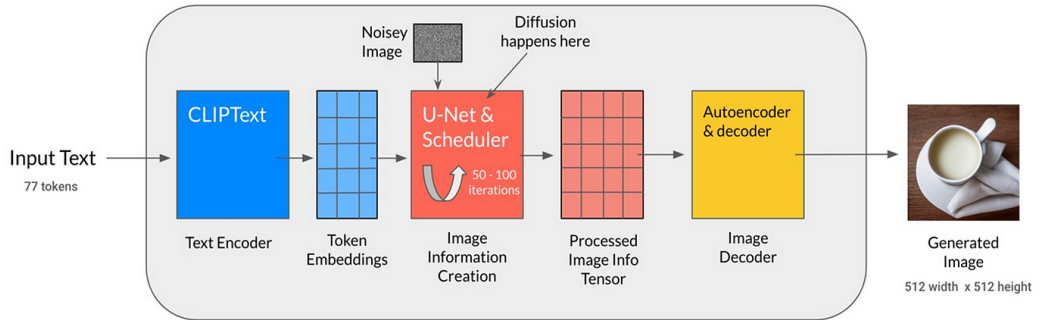


Imagen adds a pooled text embedding to the timestep embedding; it also cross-attends to the full text embedding sequence, which this diagram does not show (source: R. O'Connor, "How Imagen Actually Works", AssemblyAI, 2022).



[Source: <https://cdn.mos.cms.futurecdn.net/7zoGtryY2HfKDUg2pz8aZD.jpg>]

Stable Diffusion: An open-source diffusion model that generates high-quality images from text, designed for efficiency and accessibility.



Stable Diffusion architecture (third-party illustration, after J. Alammari, "The Illustrated Stable Diffusion", 2022).



[Source: stablediffusion-online.com, "Astronaut Riding a Horse" (SDXL) sample. The original link no longer resolves.]

Diffusion Model Architecture:

- ▶ **U-Net backbone:** The core neural network used for denoising, featuring an encoder-decoder structure with skip connections.
- ▶ **Text encoder:** Converts text prompts into embeddings. Common choices include T5 or CLIP Text Encoder.
- ▶ **Cross-attention:** Mechanism to inject text conditioning into the image generation process, allowing the model to align image features with textual information.

Comparative claims about “quality” and “prompt fidelity” rest on two measurements:

- ▶ **FID (Fréchet Inception Distance):** Fits a Gaussian to the Inception features of a set of real images and another to those of a set of generated images, then measures the Fréchet distance between them. Lower is better. It penalises unrealistic samples *and* lost variety, so it scores a set, never a single image (Heusel et al., NeurIPS 2017).
- ▶ **CLIPScore:** The cosine similarity between the CLIP embedding of the generated image and the CLIP embedding of the prompt. Higher means the image matches what was asked for (Hessel et al., EMNLP 2021).

The two pull against each other: turning the guidance scale up raises CLIPScore and, past a point, worsens FID. Papers therefore report a curve over guidance scales rather than one number.

Text-Conditioned Generation

► Concatenation of Embeddings:

- Combine text and image embeddings before feeding them into the generative model.
- Simple and effective for early fusion of modalities.

► Cross-Attention in Transformer Layers:

- Use cross-attention mechanisms to allow the model to focus on relevant parts of the text while generating images.
- Enables fine-grained alignment between text and image features.

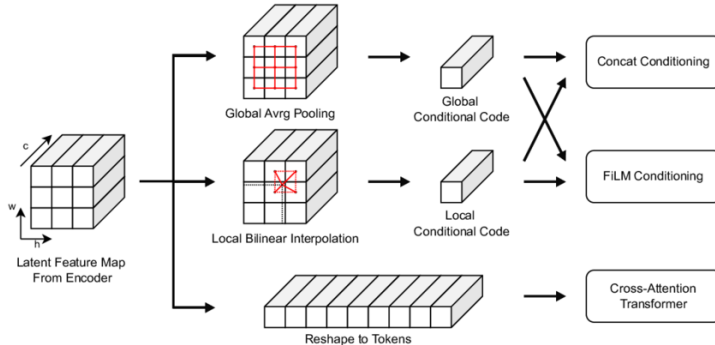
► FiLM (Feature-wise Linear Modulation; Perez et al., AAAI 2018):

- The conditioning code predicts a per-channel scale and shift applied to the decoder's features.
- Cheap, and it modulates existing channels instead of appending new ones.

► Classifier-Free Conditioning:

- Train the model with and without conditioning information.
- Allows flexible control over the influence of text during generation.

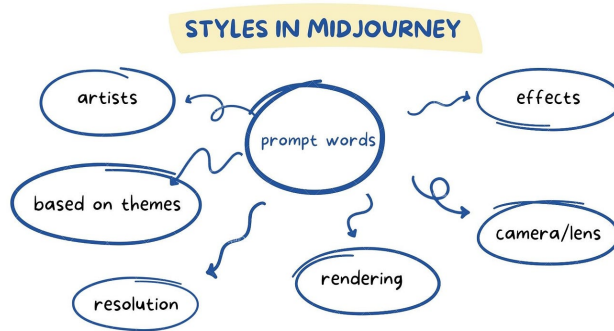
Text Conditioning Strategies (cont.)



Three ways to inject a conditioning code into a decoder: concatenation, FiLM and cross-attention. Drawn for a code from an image encoder; a text embedding is injected the same three ways (Gromniak, Magg & Wermter, ICANN 2023, Fig. 2).

- ▶ Crafting effective prompts is crucial for controlling the output of text-to-image models.
- ▶ Well-designed prompts can improve image quality, relevance, and adherence to user intent.
- ▶ Key aspects:
 - **Style Control:** Specify artistic styles (e.g., “in the style of Van Gogh”).
 - **Object Presence:** Clearly mention desired objects and their attributes.
 - **Layout:** Indicate spatial relationships (e.g., “a cat sitting on a chair”).

- ▶ **Midjourney:** "A futuristic cityscape at sunset, vibrant colors, ultra-detailed, digital art"
- ▶ **DALL-E:** "An armchair in the shape of an avocado"
- ▶ **SDXL:** "A portrait of a medieval knight, oil painting, dramatic lighting"
- ▶ Experimenting with prompt phrasing can lead to diverse and creative outputs.



An advanced guide to writing prompts for Midjourney. [Source:
https://miro.medium.com/v2/resize:fit:1400/1*4NmpCBpI8N3p1oYiknVZGA.jpeg]

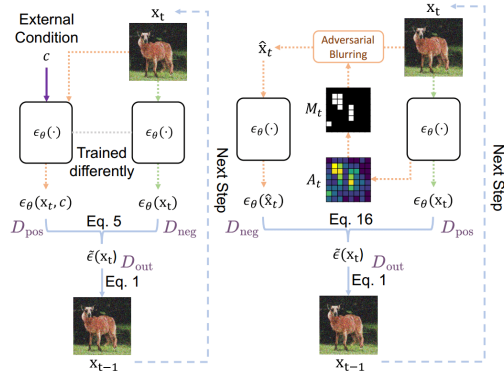
- ▶ **Ambiguity in Natural Language:** Text descriptions can be vague or open to multiple interpretations, making it difficult for models to generate the intended image.
- ▶ **Alignment with Low-Level Image Details:** Ensuring that fine-grained details in the image match the textual description remains a significant challenge.
- ▶ **Long Prompts and Hallucinations:** Handling lengthy or complex prompts can lead to hallucinated content or loss of important details in the generated images.

Classifier-Free Guidance (CFG)

- ▶ A technique (Ho & Salimans, 2022) to improve the quality of text-to-image generation by controlling the influence of text prompts.
- ▶ It trades sample diversity for closer alignment with the prompt: the guidance scale sets where on that trade-off you sit.
- ▶ CFG is particularly useful in diffusion models, enhancing the coherence between text and generated images.

- ▶ **Trades diversity for fidelity:** Adjusts the balance between mode coverage, how varied the generated set is, and how faithfully each sample follows the text prompt (Ho & Salimans, 2022).
- ▶ **No external classifier needed:** Removes the requirement for a separate classifier to guide the generation process.
- ▶ **Score extrapolation:** Extrapolates along the direction from unconditional to conditional outputs; a weight $w > 1$ pushes past the conditional score.

Key Properties of Classifier-Free Guidance (cont.)



(a) Classifier-free guidance

(b) Self-attention guidance

(a) Classifier-free guidance compared with (b) self-attention guidance (adapted from Hong et al., ICCV 2023, Fig. 2, with added $D_{\text{pos}}/D_{\text{neg}}/D_{\text{out}}$ annotations).

How does it work?

- ▶ The model generates images based on both the text prompt and random noise.
- ▶ By adjusting the guidance scale, users can control how much the text influences the final image.
- ▶ Higher guidance scales lead to images that are more closely aligned with the text, while lower scales allow for more creative freedom.

- ▶ **Duplicate forward pass:** For each input, perform two forward passes through the model:
 - One with the conditioning (e.g., text prompt)
 - One without conditioning (unconditional)
- ▶ **Combine outputs:** The final score is computed as:

$$S = S_{\text{uncond}} + w \cdot (S_{\text{cond}} - S_{\text{uncond}})$$

where:

- s_{cond} is the model output with conditioning
 - s_{uncond} is the model output without conditioning
 - w is the guidance scale parameter
- ▶ **Effect of w :** Controls the strength of guidance. Higher w enforces stronger adherence to the prompt.

- ▶ **Simpler:** Eliminates the separately trained classifier, at the cost of two forward passes ($\approx 2\times$ sampling compute) per denoising step.
- ▶ **Effective for text-guided diffusion:** Provides strong alignment between generated images and text prompts in diffusion models.
- ▶ **Foundation of modern T2I systems:** Forms the basis of Stable Diffusion XL (SDXL) and other state-of-the-art text-to-image generation systems.

Vision and Text Integration: **Summary**

- ▶ Captioning lacks true understanding
- ▶ Multimodal alignment is shallow
- ▶ Hallucinations in T2I models
- ▶ Dataset bias and fairness issues
- ▶ High computational cost

- ▶ Vision-language reasoning (e.g., VQA with logic)
- ▶ 3D scene and video generation from text
- ▶ Interactive multimodal agents
- ▶ Multilingual and cultural alignment
- ▶ Personalized generation

- ▶ Vision and text integration underpins captioning, retrieval and image generation
- ▶ Techniques include captioning, CLIP, T2I, CFG
- ▶ Rapid evolution of generative multimodal models
- ▶ Ongoing challenges and exciting frontiers

Vision and Text Integration: **References**

- [1] Vinyals, O., Toshev, A., Bengio, S., & Erhan, D.
Show and Tell: A Neural Image Caption Generator.
In *CVPR*, 2015.
<https://arxiv.org/abs/1411.4555>

- [2] Xu, K., Ba, J., Kiros, R., Cho, K., Courville, A., Salakhutdinov, R., Zemel, R., & Bengio, Y.
Show, Attend and Tell: Neural Image Caption Generation with Visual Attention.
In *ICML*, 2015.
<https://arxiv.org/abs/1502.03044>

- [3] Liu, Y., Guo, Y., & Lew, M. S.

What Convnets Make for Image Captioning?

In *MultiMedia Modeling (MMM)*, pages 416–428, 2017.

https://doi.org/10.1007/978-3-319-51811-4_34

- [4] Zhang, H., Xu, T., Li, H., Zhang, S., Wang, X., Huang, X., & Metaxas, D.

StackGAN: Text to Photo-realistic Image Synthesis with Stacked Generative Adversarial Networks.

In *ICCV*, 2017.

<https://arxiv.org/abs/1612.03242>

- [5] Krishna, R., Hata, K., Ren, F., Fei-Fei, L., & Niebles, J. C.

Dense-Captioning Events in Videos.

In *ICCV*, 2017.

<https://arxiv.org/abs/1705.00754>

- [6] Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S.

GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium (introduces the Fréchet Inception Distance).

In *NeurIPS*, 2017.

<https://arxiv.org/abs/1706.08500>

- [7] Xu, T., Zhang, P., Huang, Q., Zhang, H., Gan, Z., Huang, X., & He, X.
AttnGAN: Fine-Grained Text to Image Generation with Attentional Generative Adversarial Networks.
In *CVPR*, 2018.
<https://arxiv.org/abs/1711.10485>
- [8] Perez, E., Strub, F., de Vries, H., Dumoulin, V., & Courville, A.
FiLM: Visual Reasoning with a General Conditioning Layer.
In *AAAI*, 2018.
<https://arxiv.org/abs/1709.07871>

- [9] Pei, W., Zhang, J., Wang, X., Ke, L., Shen, X., & Tai, Y.-W.

Memory-Attended Recurrent Network for Video Captioning.

In *CVPR*, 2019.

<https://arxiv.org/abs/1905.03966>

- [10] Li, X., Yin, X., Li, C., Zhang, P., Hu, X., Zhang, L., Wang, L., Hu, H., Dong, L., Wei, F., Choi, Y., & Gao, J.

Oscar: Object-Semantics Aligned Pre-training for Vision-Language Tasks.

In *ECCV*, 2020.

<https://arxiv.org/abs/2004.06165>

- [11] Pan, B., Cai, H., Huang, D.-A., Lee, K.-H., Gaidon, A., Adeli, E., & Niebles, J. C. Spatio-Temporal Graph for Video Captioning with Knowledge Distillation. In *CVPR*, 2020.

<https://arxiv.org/abs/2003.13942>

- [12] Lee, S., & Kim, I. DVC-Net: A Deep Neural Network Model for Dense Video Captioning. In *IET Computer Vision*, 15(1), pages 12–23, 2021.

<https://doi.org/10.1049/cvi2.12013>

- [13] Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I.

Learning Transferable Visual Models From Natural Language Supervision (CLIP).

In *ICML*, 2021.

<https://arxiv.org/abs/2103.00020>

- [14] Jia, C., Yang, Y., Xia, Y., Chen, Y.-T., Parekh, Z., Pham, H., Le, Q. V., Sung, Y., Li, Z., & Duerig, T.

Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision (ALIGN).

In *ICML*, 2021.

<https://arxiv.org/abs/2102.05918>

- [15] Zhang, P., Li, X., Hu, X., Yang, J., Zhang, L., Wang, L., Choi, Y., & Gao, J.
VinVL: Revisiting Visual Representations in Vision-Language Models.
In *CVPR*, 2021.

<https://arxiv.org/abs/2101.00529>

- [16] Hessel, J., Holtzman, A., Forbes, M., Le Bras, R., & Choi, Y.
CLIPScore: A Reference-free Evaluation Metric for Image Captioning.
In *EMNLP*, 2021.

<https://arxiv.org/abs/2104.08718>

- [17] Mokady, R., Hertz, A., & Bermano, A. H.
ClipCap: CLIP Prefix for Image Captioning.
arXiv preprint, 2021.

<https://arxiv.org/abs/2111.09734>

- [18] Li, J., Li, D., Xiong, C., & Hoi, S.

BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation.

In *ICML*, 2022.

<https://arxiv.org/abs/2201.12086>

- [19] Ramesh, A., Dhariwal, P., Nichol, A., Chu, C., & Chen, M.
Hierarchical Text-Conditional Image Generation with CLIP Latents (DALL-E 2).
arXiv preprint, 2022.
<https://arxiv.org/abs/2204.06125>
- [20] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B.
High-Resolution Image Synthesis with Latent Diffusion Models (Stable Diffusion).
In *CVPR*, 2022.
<https://arxiv.org/abs/2112.10752>

- [21] Saharia, C., Chan, W., Saxena, S., Li, L., Whang, J., Denton, E., Ghasemipour, S. K. S., Karagol Ayan, B., Mahdavi, S. S., Gontijo Lopes, R., Salimans, T., Ho, J., Fleet, D. J., & Norouzi, M.

Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding (Imagen).

In *NeurIPS*, 2022.

<https://arxiv.org/abs/2205.11487>

- [22] Ho, J., & Salimans, T.

Classifier-Free Diffusion Guidance.

arXiv preprint arXiv:2207.12598, 2022. A short version appeared at the *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*.

<https://arxiv.org/abs/2207.12598>

[23] Gromniak, M., Magg, S., & Wermter, S.

Neural Field Conditioning Strategies for 2D Semantic Segmentation.

In *ICANN*, 2023.

<https://arxiv.org/abs/2304.14371>

[24] Hong, S., Lee, G., Jang, W., & Kim, S.

Improving Sample Quality of Diffusion Models Using Self-Attention Guidance.

In *ICCV*, 2023.

<https://arxiv.org/abs/2210.00939>

- [25] Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., & Rombach, R.

SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis.

arXiv preprint, 2023.

<https://arxiv.org/abs/2307.01952>

- [26] Midjourney Documentation.

<https://docs.midjourney.com/>